psystats: Basic statistics and biostatistics for psychologists in Python

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Summary

`psystats` is a Python package that provides the routine statistical procedures used in psychological research through a small, R-like functional interface. It covers descriptive statistics and grouped "Table 1" summaries with automatic selection of the appropriate significance test, common inferential tests reported with their effect sizes (Cohen's *d*, η^2, Cramér's *V*), linear and logistic regression, and epidemiological risk measures (relative risk, odds ratio, and the attributable-risk family). Every procedure returns a lightweight result object that both prints an R-style summary and can be rendered into APA-7 text or tables by the companion package `psyreport`. `psystats` is the foundational component of the **ANOVA Methods** ecosystem, which also includes `psymetrics` (reliability, exploratory and confirmatory factor analysis) and `psyreport` (APA reporting).

Statement of need

Psychologists conducting quantitative work rely heavily on a handful of R packages that pair a concise interface with output shaped for reporting: `arsenal` and `tableone` for group-comparison tables (Heinzen et al., 2021; Yoshida & Bartel, 2022), `epitools` for risk measures (Aragon, 2020), and `apaTables` for APA-formatted output (Stanley, 2021). The Python scientific stack provides the underlying estimation machinery — `pandas` (McKinney, 2010), `scipy` (Virtanen et al., 2020), and `statsmodels` (Seabold & Perktold, 2010) — but exposes it through general-purpose interfaces that require substantial boilerplate to reach a publication-ready table, and offers no direct equivalent to the "describe a data frame, get an APA table" workflow that makes the R tools attractive to applied researchers.

`psystats` closes that gap. It wraps the established estimators behind verbs that mirror their R analogues (for example `table1(df, group="condition")`, `riskratio(df, exposure=..., outcome=...)`, `alpha(items)`), so that a researcher already familiar with R incurs little translation cost, while a Python-first user obtains correct effect sizes, confidence intervals, and automatic test selection without assembling them by hand. Because results are decoupled from their presentation, the same computation feeds an interactive summary, a LaTeX table, or a Word document without recomputation. The package is aimed at researchers, students, and instructors in psychology and allied fields who work in Python and need trustworthy, reportable output for everyday analyses.

Functionality

`psystats` groups its functions into four areas. Descriptives and tables: `describe`, `freq`, `corr_matrix` (a correlation matrix with significance stars), and `table1`, which chooses a Welch *t*-test, one-way ANOVA, Mann–Whitney, Kruskal–Wallis, χ^2, or Fisher's exact test per variable according to its type and distribution. Inferential tests with effect sizes: `ttest`, `anova`, and `chisq`. Regression: `linreg`, reporting standardized coefficients and model fit, and `logreg`, reporting odds ratios with confidence intervals. Risk factors: `riskra-

`tio`, `oddsratio`, and `attributable_risk`, which returns the risk difference with its confidence interval, the attributable risk percent, the population attributable risk and its percent, and the number needed to treat.

Quality control

All estimates are validated in an automated test suite against independent references — `scipy` and `statsmodels` for test statistics, confidence intervals, and regression coefficients, and closed-form identities for the effect sizes — so that reported values match established implementations rather than the package's own internals.

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References

Aragon, T. J. (2020). *Epitools: Epidemiology tools*. https://CRAN.R-project.org/package=epitools

Heinzen, E., Sinnwell, J., Atkinson, E., Gunderson, T., & Dougherty, G. (2021). *Arsenal: An arsenal of R functions for large-scale statistical summaries*. https://CRAN.R-project.org/package=arsenal

McKinney, W. (2010). Data structures for statistical computing in Python. *Proceedings of the 9th Python in Science Conference*, 56–61. https://doi.org/10.25080/Majora-92bf1922-00a

Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with Python. *Proceedings of the 9th Python in Science Conference*, 92–96. https://doi.org/10.25080/Majora-92bf1922-011

Stanley, D. (2021). *apaTables: Create american psychological association (APA) style tables*. https://CRAN.R-project.org/package=apaTables

Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, *17*, 261–272. https://doi.org/10.1038/s41592-019-0686-2

Yoshida, K., & Bartel, A. (2022). *Tableone: Create 'table 1' to describe baseline characteristics*. https://CRAN.R-project.org/package=tableone

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Software repository: https://github.com/anovabr/anova-methods · License: MIT · Archived releases and the citable DOI are listed in `CITATION.cff`.